PROJECT REQUIREMENTS

Active Directory Attack Lab: Kerberoasting Exposure Review and GPO Hardening

1. Hardware Requirements

- 64-bit computer with virtualization support (Intel VT-x/AMD-V).
- Recommended: 16 GB RAM or more for a Windows Server VM and one or more Windows client VMs.
- At least 100 GB free disk space for VM disks and snapshots.
- Network adapter capable of host-only or isolated virtual networking.

2. Software Requirements

- Windows Server evaluation or licensed edition for the domain controller.
- Windows client VM(s), such as Windows 10/11, as compatible with the chosen lab setup.
- Hypervisor such as Hyper-V, VirtualBox, or VMware Workstation.
- Python 3.10 or newer to run the included audit utility.
- Text editor or IDE (VS Code, PyCharm, or equivalent).

3. Python Dependencies

No third-party Python packages are required. The script uses only the standard library: argparse, json, pathlib, and typing.

4. Input Data Requirements

The audit script reads a JSON inventory with a domain name, service account records, SPN lists, password age in days, supported encryption types, and selected GPO values. A synthetic sample is provided as **sample_inventory.json**.

5. Security and Lab Isolation

- Use an isolated, authorized virtual environment only.
- Do not connect the lab to production domain controllers or production networks.
- Use synthetic accounts and non-sensitive test data.
- Take VM snapshots before configuration changes.
- Do not store real passwords, password hashes, tickets, or secrets in the inventory or report.

6. Validation Criteria

- The Python script accepts the sample JSON and creates a JSON audit report.
- SPN-bearing accounts and stale service-account password indicators are identified for review.
- Selected GPO settings are classified as REVIEW, NOT_VERIFIED, or passing (not listed as a finding).
- Findings include practical remediation guidance and known limitations.